

1. (3 pontos) Compare the New Keynesian framework with both the New Classical and original Keynesian frameworks. Organize your comparison along the following dimensions:
 - i. price adjustment, expectations, and the short-run effects of monetary policy;
 - ii. market structure and firms' price-setting behavior;
 - iii. the construction of aggregate demand and the monetary-policy instrument;
 - iv. the use of microeconomic foundations and welfare analysis.

For each dimension, discuss the empirical motivation for the New Keynesian formulation and its implications for the structure and predictions of the model.

2. (3 pontos) Using separate AS–AD diagrams, analyze the following changes:
 - a) a temporary increase in firms' desired markups;
 - b) a reduction in the nominal interest rate by the Central Bank.

In each case, identify which curve shifts, indicate the direction of the shift, and describe the effects on output and the price level. Explain the economic intuition behind the movement of the curve and the resulting equilibrium.

3. (2 pontos) Carefully distinguish between the natural level of output, y_n , and the efficient level of output, y_e . Why can these two levels differ in an economy with monopolistic competition? Explain why a temporary increase in firms' desired markups changes y_n but not y_e .
4. (2 pontos) The microeconomic foundations of the New Keynesian model make it possible to evaluate monetary policy in terms of household welfare.

Consider the following loss function in conjunction with the AS-AD dynamics seen in Benigno (2015)

$$\begin{aligned} \min_{y,p} \mathcal{L} &= \phi_y(y - y^*)^2 + \phi_p(p - p^e)^2, & \phi_y, \phi_p > 0, \\ \text{s.t.} & \\ p - p^e &= \kappa(y - y_n). \end{aligned}$$

Interpret the shape of the loss function and analyse what it implies for the behavior of a central bank. Also, explain why this problem is subjected to the AS condition and what does that mean in terms of economic policy